Predator-prey systems

We've seen how differential equations can be used to model the size of a population over time. Of course, populations don't tend to exist in isolation; they usually interact with other populations, either as predator or prey.

Populations as a system

For that reason, it's common to consider two populations as a system, and model their interaction with a system of differential equations. In an environment with two populations x and y, where dx/dt models population x over time t and dy/dt models population y over time t, we can always classify the system as cooperative, competitive, or predator-prey.

- In **cooperative** systems, both populations are increasing in size (the mixed xy terms are both positive).

$$\frac{dx}{dt} = ax + bxy \qquad \frac{dy}{dt} = cy + dxy$$

- In **competitive** systems, both populations are decreasing in size (the mixed xy terms are both negative).

$$\frac{dx}{dt} = ax - bxy \qquad \frac{dy}{dt} = cy - dxy$$

- In **predator-prey** systems, the size of one population is increasing, while the size of the other population is decreasing (one mixed xy term is positive while the other is negative).

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Population x is increasing while population y is decreasing

$$\frac{dx}{dt} = ax + bxy \qquad \frac{dy}{dt} = cy - dxy$$

or

Population y is increasing while population x is decreasing

$$\frac{dx}{dt} = ax - bxy \qquad \frac{dy}{dt} = cy + dxy$$

Sometimes one or both differential equations in the system will contain higher degree terms, like the bx^2 term in the dx/dt equation in the following system.

$$\frac{dx}{dt} = ax + bx^2 + cxy$$

$$\frac{dy}{dt} = fy - gxy$$

When the differential equations modeling the populations don't include a higher degree term, it means both populations are only affected by one another. But when a higher degree term is included, like the bx^2 above, it means that population is affected by not only the other population in the system, but also by another factor, like carrying capacity.

Stable solutions of the system

The system of differential equations modeling two interacting populations will have predictable equilibrium solutions.

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1. The system will have a stable equilibrium solution $(0,0)$, where both populations x and y are at 0 and will remain at 0 indefinitely.

2. If the system has an equilibrium solution at $(a,0)$, it means population x is stable at size a, while population y is at 0.

3. If the system has an equilibrium solution at $(0,b)$, it means population y is stable at size b, while population x is at 0.

4. The most interesting solution is an equilibrium solution at (a,b), where population x has size a, which is supporting in perfect balance population y at size b, and vice versa.

Let's do an example where we solve a population system for its stable and unstable solutions.

Example

Does the system of lions and zebras represent a cooperative, competitive, or predator-prey system? What are the equilibrium solutions of the system and what do they tell us about the balance of the populations?

$$\frac{dL}{dt} = -0.5L + 0.0001LZ$$

$$\frac{dZ}{dt} = 2Z - 0.0002Z^2 - 0.01LZ$$

The mixed LZ term $0.0001LZ$ in the differential equation for lions dL/dt is positive, while the mixed LZ term $-0.01LZ$ in the differential equation for

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zebras dZ/dt is negative, which means this is a predator-prey system in which the lions are the predators and the zebras are the prey.

To find equilibrium solutions, we'll factor both equations to get

$$\frac{dL}{dt} = -0.5L + 0.0001LZ$$

$$\frac{dL}{dt} = 0.0001L(-5{,}000 + Z)$$

$$\frac{dL}{dt} = 0.0001L(Z - 5{,}000)$$

and

$$\frac{dZ}{dt} = 2Z - 0.0002Z^2 - 0.01LZ$$

$$\frac{dZ}{dt} = 0.0002Z(10{,}000 - Z - 50L)$$

Setting both equations equal to 0 gives

$$0.0001L(Z - 5{,}000) = 0$$

$0.0001L = 0$ and $Z - 5{,}000 = 0$

$L = 0$ and $Z = 5{,}000$

and

$$0.0002Z(10{,}000 - Z - 50L) = 0$$

$0.0002Z = 0$ and $10{,}000 - Z - 50L = 0$

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$$Z = 0 \text{ and } Z + 50L = 10{,}000$$

We need to test both solutions from the first equation with both solutions from the second equation. So if we pair $L = 0$ with both $Z = 0$ and $Z + 50L = 10{,}000$, and then in addition we pair $Z = 5{,}000$ with both $Z = 0$ and $Z + 50L = 10{,}000$, we get four combinations,

$$L = 0 \text{ and } Z = 0$$

$$L = 0 \text{ and } Z + 50L = 10{,}000$$

$$Z = 5{,}000 \text{ and } Z = 0$$

$$Z = 5{,}000 \text{ and } Z + 50L = 10{,}000$$

We're looking for each pair to generate an equilibrium solution (L, Z). The third pair, $Z = 5{,}000$ and $Z = 0$, doesn't include a value for L, so we can eliminate that pair completely and focus on just the other three.

$$L = 0 \text{ and } Z = 0$$

$$L = 0 \text{ and } Z + 50L = 10{,}000$$

$$Z = 5{,}000 \text{ and } Z + 50L = 10{,}000$$

The first pair gives the trivial solution $(L, Z) = (0,0)$ where both populations are at 0. To solve the second equation pair, we'll substitute $L = 0$ into $Z + 50L = 10{,}000$,

$$Z + 50(0) = 10{,}000$$

$$Z = 10{,}000$$

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and to solve the third equation pair, we'll substitute $Z = 5{,}000$ into $Z + 50L = 10{,}000$,

$$5{,}000 + 50L = 10{,}000$$

$$50L = 5{,}000$$

$$L = 100$$

The solutions of the system are therefore

- $(L, Z) = (0,\ 0)$, where both populations are 0 and will remain there
- $(L, Z) = (0,\ 10{,}000)$, where the zebra population is stable at $10{,}000$ while the lion population is at 0.
- $(L, Z) = (100,\ 5{,}000)$, where the system is balanced, with 100 lions supporting in balance $5{,}000$ zebras, and vice versa.

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